BUILDING MODEL DESCRIPTION - ARCHETYPE QC

CI_EDU_SchoolSec-clim_Large_Pre1970

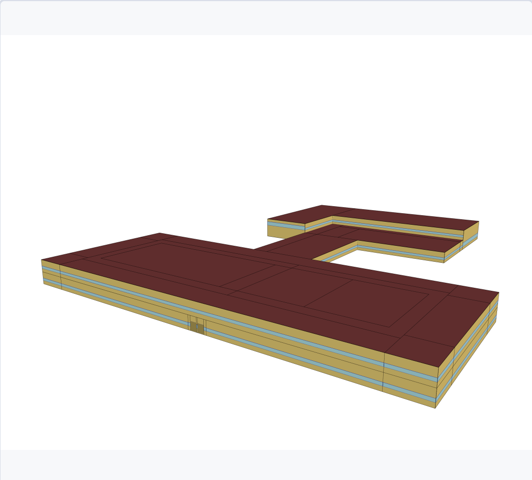
VERSION | DATE
2026-06-15

SECTION 1 — MODEL DESCRIPTION

BUILDING DESCRIPTION

Sector	Commercial-Institutional
Building type	Education
Building subtype	Secondary school, with cooling - Large
Vintage	Before 1970

3D MODEL



Three-dimensional geometric representation of the building model used for energy simulation and performance analysis.

BUILDING CHARACTERISTICS

Floor Area [m²]	17204
Climate zone	

HVAC SYSTEM

Cooling system	Central Electric Heat Pump System
Heating system	Electric Boiler with hydronic baseboard heaters for perimeter
Ventilation system	Constant air volume with recirculation and terminal reheat
Typical COP cooling	2.60

INSULATION & AIR TIGHTNESS

Exterior walls [W/m²/K]	0.72
Interior walls [W/m²/K]	0
Roof [W/m²/K]	0.28
Slabs [W/m²/K]	1.91
Infiltration (I_design) [m³/h/m²]	2.05

GLAZING

Glazing [W/m²/K]	3.52
SHGC	0.41

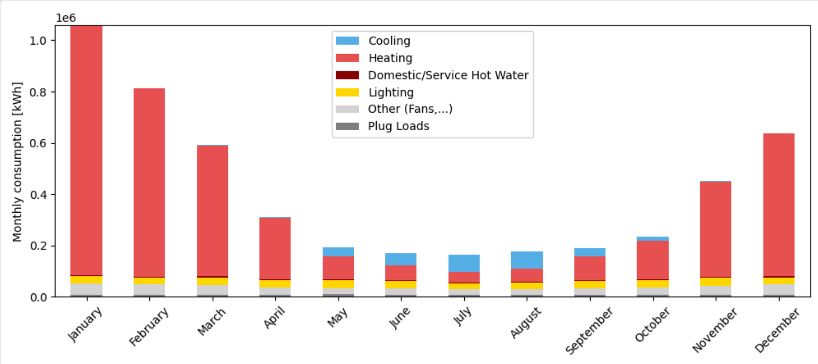
WINDOW-TO-WALL RATIO

Average [%]	19.7

ANNUAL END USES — ELECTRICITY [kWh]

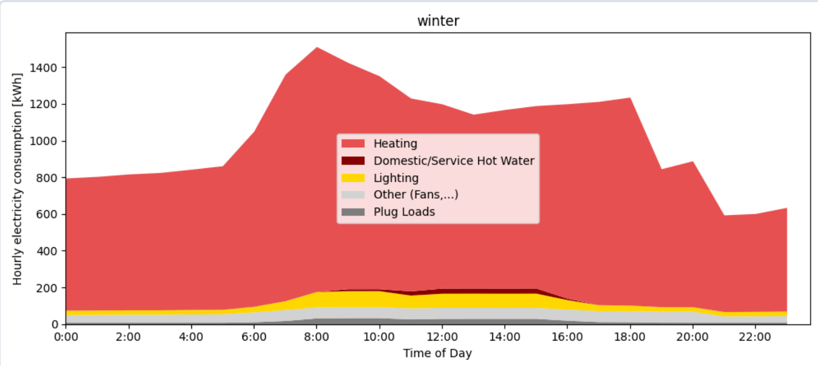
END USE	[kWh/year]
Heating	3873098
Cooling	275013
Interior Lighting	335414
Interior Equipment	132576
Fans	258235
Pumps	70809
Water Systems	44992
Total End Uses	4990137

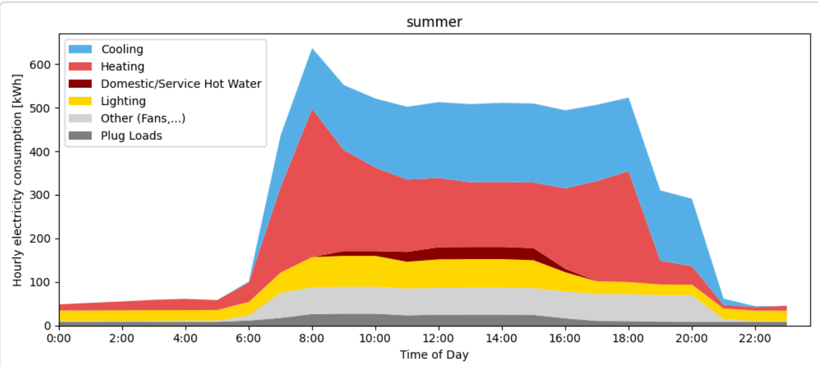
MONTHLY CONSUMPTION BY END USE



Monthly energy consumption breakdown by end use, showing the contribution of heating, cooling, lighting, domestic hot water, plug loads, and auxiliary systems throughout the year.

AVERAGE DAILY PROFILE BY USAGE





Average daily consumption profile for winter and summer, showing the contribution of cooling, lighting, domestic hot water, plug loads, and auxiliary systems throughout an average winter and summer day.

ENERGY USE INTENSITY	TOTAL CONSUMPTION	ELECTRICITY	NATURAL GAS
290 kWh/m²/yr	4990101 kWh/yr	4990137 kWh/yr	0 kWh/yr